custom_label	question_count	problem_ids	problem_titles
Backward Pointer Scan	2	58, 3856	Length of Last Word |
Binary Lifting (Sparse Table)	2	1483, 2836	Kth Ancestor of a Tre
Closure	2	2627, 2676	Debounce (JS) | Thro
Gaps and Islands (Row Number Difference)	2	1225, 1285	Report Contiguous Da
Hierholzer's Algorithm (Eulerian Path)	2	753, 2097	Cracking the Safe | V
LCA (Lowest Common Ancestor)	2	2096, 2846	Step-By-Step Directio
Single-Pass Flagging	2	2299, 3136	Strong Password Che
Single-Pass Linear Scan	2	3285, 3386	Find Indices of Stable
Two Min-Heaps (Priority Queues)	2	1942, 2462	The Number of the Sm
2D-to-1D Reduction	1	1074	Number of Submatric
8-Directional Raycasting from Move	1	1958	Check if Move is Lega
Accumulating Closure Scopes (Each timeout creates a new closure, potentially keeping old variables in memory longer)	1	2725	Interval Cancellation
Amortized Linear Scan / Precomputation	1	3255	Find the Power of K-S
Angular Sweep / Fixed Circle Centers	1	1453	Maximum Number of
Arithmetic Computation	1	2469	Convert the Temperat
Array Grouping by Set Bits	1	3011	Find if Array Can Be S
Array Indexing (Direct Access)	1	1603	Design Parking Syste
Array Length Property (`this.length`)	1	2619	Array Prototype Last
Array modification tracking (snapshotting the entire array state at each simulated step)	1	2702	Minimum Operations
Backwards Traversal (Target to Start)	1	3724	Minimum Operations
Base 3 Representation (Ternary Logic)	1	1780	Check if Number is a
Bi-directional Linear Scan with Last-Seen Map. O(N	1	3488	Closest Equal Elemen
Binary Exponentiation	1	50	Pow(x, n)
Binary Exponentiation (Fast Power)	1	1922	Count Good Numbers
Binary Offset Mapping	1	1256	Encode Number
Blocking Main Thread (Visualizing the CPU stuck in a loop waiting for the async task)	1	2776	Convert Callback to P
Boolean Array	1	2682	Find the Losers of the
Boolean Indexing: df[df['id'] == 101]	1	2880	Select Data (Filtering
Boundary Contraction	1	59	Spiral Matrix II
Boundary Shrinking	1	54	Spiral Matrix
Brainteaser / Logical Deduction	1	1332	Remove Palindromic S
Brute Force (Small Constraints)	1	2932	Maximum Strong Pair
Brute Force / Constraint Optimization	1	1534	Count Good Triplets
Brute Force Path Memory (Storing every conceivable route configuration simultaneously)	1	2714	Find Shortest Path wi
Brute Force with Early Exit (Small Constraints)	1	3345	Smallest Divisible Dig
Buffer Scan (Visualizing the database engine reading the text column repeatedly for each word)	1	2738	Count Occurrences in
Buffered Chunk Processing with Bound Checking	1	157	Read N Characters Gi
COUNT(DISTINCT customer_id)	1	2082	The Number of Rich C
Chained Operations: df.loc[...].sort_values(...)[['name']]	1	2891	Method Chaining
Chebyshev Distance (Max of ΔX, ΔY)	1	1266	Minimum Time Visitin
Closure / Higher-Order Function	1	2666	Allow One Function C
Column-Wise Linear Scan	1	2639	Find the Width of Col
Column-Wise Max Pre-calculation	1	3033	Modify the Matrix
Combinatorial Construction. Generate palindromes directly by mirroring the "first half" of the binary bits	1	3677	Count Binary Palindro
Combinatorial Row Storage (Visualizing all row combinations in memory)	1	2732	Find a Good Subset o
Compare Current vs Next	1	13	Roman to Integer
Concurrency: 3 Semaphores	1	1116	Print Zero Even Odd
Concurrency: Mutexes / Semaphores / Promises	1	1114	Print in Order
Concurrency: Two Alternating Semaphores	1	1115	Print FooBar Alternate
Condition Variables / Semaphores	1	1188	Design Bounded Bloc
Contribution of Each Character (Index tracking)	1	828	Count Unique Charac
Convex Hull	1	3588	Find Maximum Area o
Coordinate Mapping (Direct Indexing)	1	2596	Check Knight Tour Co
Coordinate Mapping (Precomputation)	1	3242	Design Neighbor Sum
Coordinate Min-Max Comparison	1	3047	Find the Largest Area
Coordinate Rotation (Chebyshev Transformation)	1	3102	Minimize Manhattan D
Corner Count Set	1	391	Perfect Rectangle
Cross Product (Orientation) For every three consecutive points, calculate the cross product. All cross products must have the same sign (all positive or all negative)	1	469	Convex Polygon
Custom Comparator / Partitioning	1	937	Reorder Data in Log F

Cycle Decomposition (Permutation Theory)	1	3863	Minimum Operations
Date Filtering	1	2993	Friday Purchases I
Days Since Epoch (1971) Calculation	1	1360	Number of Days Betw
Direct Comparison / O(1) Access	1	2236	Root Equals Sum of C
Direct Index Mapping (One-Pass)	1	2075	Decode the Slanted C
Direct Indexing Compare `words[i].charAt(j)` with `words[j].charAt(i)` with boundary checks	1	422	Valid Word Square
Direct Integer Accumulation	1	1945	Sum of Digits of Strin
Direct Linear Scan with Set Lookup	1	2586	Count the Number of
Direct Mapping	1	953	Verifying an Alien Dic
Direct SQL WHERE Clause	1	620	Not Boring Movies (S
Divisor Iteration	1	3138	Minimum Length of A
Double In-place Reversal (Reverse Whole	1	186	Reverse Words in a S
Double Logarithmic Loop (Logarithmic Bounds)	1	970	Powerful Integers
Double Loop with Square Root Check	1	1925	Count Square Sum Tr
Dual-Map Bijection or Pattern Normalization	1	890	Find and Replace Patt
Dynamic Prefix Maximum & Max-Difference Tracking	1	2874	Maximum Value of ar
Dynamic SQL / Information Schema	1	2253	Dynamic Unpivoting o
Early Exit Iterator	1	1961	Check If String Is a P
Early-Exit Linear Comparison	1	2828	Check if a String Is a
Edge-Centric	1	2242	Maximum Score of a
Enumeration by Zero-Count (O(N√N))	1	3234	Count Substrings Wit
Execution Context Overhead (Visualizing the engine creating and destroying complex scopes for every iteration)	1	2804	Array Prototype ForEa
Exhaustive Heatmap Search (Brute Force)	1	1620	Coordinate With Maxi
Exponential State Memory (Visualizing the storage of all valid subarray combinations)	1	2771	Longest Non-decreas
Exponential State Search (Visualizing all possible combinations of powers being summed)	1	2787	Ways to Express an I
Fast Fourier Transform (FFT) or Karatsuba	1	3549	Multiply Two Polynom
Fast and Slow Pointers	1	2095	Delete the Middle Nod
Filtered Select with Null Handling	1	584	Find Customer Refere
Find Dip, Swap, Reverse	1	31	Next Permutation
Finding Bound Indices (Last 1, First 3)	1	3431	Minimum Unlocked In
First/Last Index Pre-computation	1	1930	Unique Length-3 Palin
Fisher-Yates Shuffle Algorithm	1	384	Shuffle an Array
Fixed-Window Sliding (Multiple Passes)	1	2067	Number of Equal Cou
Floating Closure Memory (The callback and its lexical environment stay in memory until the timer naturally finishes)	1	2715	Timeout Cancellation
Function with Local Variable Offset (N-1) and LIMIT	1	177	Nth Highest Salary
Function.prototype.apply / Symbol Key	1	2693	Call Function with Cu
Generators (`yield`)	1	2648	Generate Fibonacci S
Global Bounding Box Scan	1	2249	Count Lattice Points I
Global Sum Maintenance (Delta Tracking)	1	985	Sum of Even Number
Group Consecutive Counts	1	696	Count Binary Substrin
Grouping by Vowel Count	1	3775	Reverse Words With S
IN Clause with Multi-Column Key	1	1174	Immediate Food Deliv
In-Degree Accumulator (Array)	1	2374	Node With Highest Ed
In-Order Flattening	1	1382	Balance a Binary Sea
In-Order Traversal (Sorted Property)	1	783	Minimum Distance Be
In-Order Traversal (natively sorted)	1	530	Minimum Absolute Di
In-Place "Leapfrog" Reversal	1	92	Reverse Linked List II
In-Place Group Iteration & Re-linking	1	2074	Reverse Nodes in Eve
In-Place In-order Maintain a `first` and `last` pointer. Connect `last.right = current` and `current.left = last` during traversal	1	426	Convert BST to Sorte
In-Place Iterative Reduction	1	2293	Min Max Game
In-Place Link Manipulation	1	1474	Delete N Nodes After
In-Place Pointer Reversal	1	25	Reverse Nodes in k-G
In-Place Pointer Splicing	1	2807	Insert Greatest Comm
In-Place State Marking	1	73	Set Matrix Zeroes
In-order Traversal with In-place Re-linking	1	897	Increasing Order Sea
In-place Array Mutations (Visualizing the CPU repeatedly writing to memory for every range update)	1	2772	Apply Operations to N
In-place Array Transformation	1	544	Output Contest Match
In-place Marking For each number x, go to index |x|-1 and flip the value to negative. If already negative, it's a duplicate	1	442	Find All Duplicates in
In-place Negation For each value x, treat |x|-1 as an index and make the value at that index negative. Positives remaining are the gaps	1	448	Find All Numbers Disa
In-place array mutations (Unnecessary write operations to memory)	1	2717	Semi-Ordered Permu
In-place modification of Character Array	1	1844	Replace All Digits with

In-place row swapping within the specified submatrix boundaries	1	3643	Flip Square Submatri
Index Mapping	1	1773	Count Items Matchin
Index-as-Hash / Negation Marker	1	645	Set Mismatch
Information Theory (N-Dimensional Hypercube) Each pig represents a dimension. The number of states a pig can have is (Tests	1	458	Poor Pigs
Integer State Mutation (Visualizing the variables main and additional changing over time)	1	2739	Total Distance Travel
Intermediate Array Allocation (Visualize a potentially massive array of strings being created in memory just for a boolean check)	1	2727	Is Object Empty
Inverse Transform (Polar) Generate theta in [0, 2π] and r = R √{rand()}. The square root is crucial for uniform density	1	478	Generate Random Po
Iterative Accumulator Update	1	2626	Array Reduce Transfo
Iterative Brute Force (Optimized Range)	1	2928	Distribute Candies Ar
Iterative Cleanup	1	1694	Reformat Phone Num
Iterative Comparison	1	2264	Largest 3-Same-Digit
Iterative In-order Traversal with Early Stop	1	230	Kth Smallest Element
Iterative Mapping (New Array)	1	2635	Apply Transform Ove
Iterative New Array Push	1	2634	Filter Elements from
Iterative Pointer Manipulation (Bottom-Up Logic)	1	156	Binary Tree Upside D
Iterative Pointer Mutation (Visualizing the CPU cycle overhead for millions of small increments)	1	2757	Generate Circular Arr
Iterative Slicing (Step = Size)	1	2677	Chunk Array
JavaScript Generators & Promises	1	2650	Design Cancellable F
JavaScript Prototype Chain Traversal	1	2618	Check if Object Insta
JavaScript Proxy (`get` trap)	1	2690	Infinite Method Objec
LCP Array (Longest Common Prefix) / Z-Algorithm	1	3388	Count Beautiful Splits
LIS (Longest Increasing Subsequence)	1	1671	Minimum Number of
Lexical Scope / Closures	1	2620	Counter (JS)
Linear Algebra (System of Equations)	1	1276	Number of Burgers w
Linear Array Scanning	1	2951	Find the Peaks
Linear Boundary Comparison	1	1272	Remove Interval
Linear Count Tracker	1	3254	Find the Power of K-S
Linear Memory Scan (Visualizing the CPU reading the entire marbles array for every single instruction)	1	2766	Relocate Marbles
Linear One-Pass Update	1	2239	Find Closest Number
Linear Row Scan	1	2643	Row With Maximum
Linear Scan (Max Property Observation)	1	2419	Longest Subarray Wi
Linear Scan (Min/Max Tracking)	1	3279	Maximum Total Area
Linear Scan / Substring Search	1	2124	Check if All A's Appe
Linear Scan of Adjacent Elements	1	163	Missing Ranges
Linear Scan with State Tracker	1	2660	Determine the Winne
Linear Search Overhead (Visualizing indexOf being called repeatedly)	1	2794	Create Object from T
Linear Search Pointers (Drawing the search range for every individual query)	1	2747	Count Zero Request S
Linear Traversal (Bidirectional)	1	3263	Convert Doubly Linke
Linear loop checking `nums[i]-nums[i-1]`	1	3423	Maximum Difference
List Maintenance	1	2899	Last Visited Integers
Local Incremental Update	1	2672	Number of Adjacent
Logical Comparison of First Two Edges	1	1791	Find Center of Star G
Logical Invariance (Existence of '1')	1	2546	Apply Bitwise Operati
Longest Common Substring (LCS Variant)	1	3135	Equalize Strings by A
Look-Ahead (Caching)	1	284	Peeking Iterator
Loop pointers footprint (Draw i and j pointers constantly jumping back and forth across an array)	1	2706	Buy Two Chocolates
Lowest Common Ancestor (LCA)	1	1740	Find Distance in a Bi
Lowest Common Ancestor (LCA) with Binary Lifting	1	3515	Shortest Path in a We
Manacher's Algorithm	1	3504	Longest Palindrome A
Massive String Reallocations (Immutable strings mean creating a brand new string in memory for every deletion)	1	2716	Minimize String Leng
Maximum Flow / Bipartite Matching. Model robots as one set of nodes, walls as another, and find the maximum matching/coverage	1	3661	Maximum Walls Destr
Median	1	2967	Minimum Cost to Mak
Median Property (Independent Dimensions)	1	296	Best Meeting Point
Middle Character Extraction	1	2011	Final Value of Variabl
Min-Cost Flow / Hungarian Algorithm	1	3785	Minimum Swaps to A
Min-Max Coordinate Tracking	1	3195	Find the Minimum Ar
Min/Max Tracking	1	2148	Count Elements With
Minimal Memory (Only tracking unique category representatives)	1	2782	Number of Unique Ca
Minimal Variable State (Only index i)	1	2774	Array Upper Bound (1
Minimax / Information Theory (Heuristic Guessing)	1	843	Guess the Word
Minimum Boundary Tracking	1	598	Range Addition II

Mismatch Index Comparison	1	1790	Check if One String S
Native Pandas Constructor (Vectorized Internal Logic)	1	2877	Create a DataFrame f
Nested Char Loops	1	2194	Cells in a Range on a
Net Shift Calculation	1	1427	Perform String Shifts
Network Socket Overload (Visualizing multiple open connections consuming system file descriptors)	1	2756	Query Batching (JS)
Next Permutation Algorithm (Linear Scan)	1	556	Next Greater Element
Next Permutation Algorithm (Suffix Reversal)	1	3720	Lexicographically Sm
Next Permutation Algorithm on the first half	1	1842	Next Palindrome Usin
Next Permutation on Half-String	1	3734	Lexicographically Sm
Number Theory (Greatest Common Divisor)	1	1250	Check If It Is a Good
O(1) State Variables (Draw one sum and one i)	1	2778	Sum of Squares of Sp
O(N^2) 2D Array Allocation (Massive memory waste for large N)	1	2718	Sum of Matrix After C
ORDER BY Multiple Columns	1	2377	Sort the Olympic Tab
One-Pass Min-Max Tracking	1	1491	Average Salary Exclu
One-Pass Neighbor Comparison	1	766	Toeplitz Matrix
One-Pass Precomputation	1	3349	Adjacent Increasing S
One-Pass Running Maximum	1	915	Partition Array into D
One-Pass String Comparison	1	161	One Edit Distance
One-Pass with Toggle/Previous Tracking	1	245	Shortest Word Distan
Optimized Linear Scan (Early Exit)	1	2452	Words Within Two Edi
Parity Logic (Difference is even iff Total Sum is even). O(N)	1	3432	Count Partitions with
Parity Summation	1	3274	Check if Two Chessbo
Partition Memory Sets (Visualizing two massive separate heaps for every single partition attempt)	1	2740	Find the Value of the
Pattern Canonicalization / Hashing	1	1072	Flip Columns For Max
Pattern Categorization (6-Segment Window)	1	335	Self Crossing
Pattern Searching (Presence of "01")	1	1784	Check if Binary String
Permutation Validation (Fixed Size)	1	949	Largest Time for Give
Persistent Pointer (Amortized Analysis)	1	2424	Longest Uploaded Pre
Polynomial Class / Map of Term-to-Coefficient	1	770	Basic Calculator IV
Pre-allocation	1	1313	Decompress Run-Len
Pre-evaluated Boolean Variables mapping to strict logic combinations	1	2525	Categorize Box Accor
Pre-mapped Char-to-Row Create a Map or Array of size 26 where `row[char]` is 1, 2, or 3. Check if all chars in a word return the same row ID	1	500	Keyboard Row
Pre-processing (Index Collection)	1	3159	Find Occurrences of a
Precomputation (Prefix Products)	1	2438	Range Product Queri
Precomputation (RowithCol Sum Arrays)	1	2482	Difference Between C
Precompute "Longest Decreasing" from left and "Longest Increasing" from right for every index	1	3676	Count Bowl Subarray
Precomputed Row-Mins and Col-Maxes	1	1380	Lucky Numbers in a M
Prefix & Suffix Arrays	1	3334	Find the Maximum Fa
Prefix & Suffix Arrays (State Tracking)	1	1525	Number of Good Way
Prefix & Suffix Products (Flattened 2D to 1D)	1	2906	Construct Product Ma
Prefix Expansion / Direct String Construction	1	3324	Find the Sequence of
Prefix Manipulation	1	564	Find the Closest Palin
Prefix Max & Suffix Min	1	768	Max Chunks To Make
Prefix Max & Suffix Min Pass	1	1966	Binary Searchable Nu
Prefix Max / Suffix Max Arrays	1	3745	Maximize Expression
Prefix Max and Suffix Min Arrays	1	2012	Sum of Beauty in the
Prefix Maximum & Suffix Maximum (or Pre-calculation)	1	2873	Maximum Value of ar
Prefix Min & Suffix Min Arrays	1	2908	Minimum Sum of Mou
Prefix Min & Suffix Min Arrays (Identical to 2908)	1	2909	Minimum Sum of Mou
Prefix Product Array with Zero-Reset	1	1352	Product of the Last K
Prefix and Suffix Accumulators	1	3788	Maximum Score of a
Prefix and Suffix Counts. Count existing '0's and '1's. Inserting '0' at the start pairs with all existing '1's. Inserting '1' at the end pairs with all existing '0's	1	3628	Maximum Number of
Prefix and Suffix Precomputation Arrays	1	3817	Good Indices in a Dig
Prefix and Suffix Sums. Precalculate even/odd sums. When you "remove" an element, the suffix even/odd sums simply swap roles!	1	3634	Minimum Removals to
Prepared Statements / Dynamic SQL	1	2252	Dynamic Pivoting of a
Preprocessing (RowithCol Sums)	1	1582	Special Positions in a
Promise.allSettled Polyfill	1	2795	Parallel Execution of I
Promise.race() with Timer	1	2637	Promise Time Limit
Promises	1	2621	Sleep (JS)
Proxy	1	2691	Immutability Helper (

Range Narrowing (Iterative Intersection). O(N)	1	3468	Find the Number of C
Redundant Array Rotation Buffers (Storing the state of the array for every single shift)	1	2735	Collecting Chocolates
Redundant Decision Paths (Visualizing same wall counts being re-calculated in different branches)	1	2742	Painting the Walls
Redundant Min/Max Scans (Drawing a search arrow within every window)	1	2762	Continuous Subarrays
Resource Hierarchy / Semaphores	1	1226	The Dining Philosoph
Reverse In-Order Traversal (Right → Root → Left)	1	538	Convert BST to Great
Reverse In-order Traversal (Right -> Root -> Left)	1	1038	Binary Search Tree to
Reverse Linear Scan	1	66	Plus One
Reverse Linear Scan (Right-to-Left)	1	1299	Replace Elements wit
Reverse Traversal Start from the end of the string. Collect K characters, add a dash, and repeat. Reverse the result at the end	1	482	License Key Formattin
Revert Half the Number	1	9	Palindrome Number
Right Max Array	1	3073	Maximum Increasing
Right-Spine Insertion	1	998	Maximum Binary Tree
Right-to-Left Array Reduce	1	2629	Function Composition
Right-to-Left Monotonic Scan	1	1762	Buildings With an Oce
Row Max, Column Max, and Non-Zero Count	1	883	Projection Area of 3D
Row-by-Row Comparison Buffer (Visualizing the SQL engine re-scanning the bookings table for every single record)	1	2793	Status of Flight Ticke
Row-by-Row Memory Copy (Intermediate tables filling up Cartesian space)	1	2701	Consecutive Transact
Row-wise Iteration (Simple Scan)	1	1672	Richest Customer We
Running Maximum Comparison	1	769	Max Chunks To Make
Running Maximum Property	1	775	Global and Local Inve
SELECT DISTINCT	1	1148	Article Views I (SQL)
SQL DATE_FORMAT() function	1	1853	Convert Date Format
SQL IS NULL	1	3358	Books with NULL Rati
SQL LEAST() and GREATEST() Functions	1	1699	Number of Calls Betw
SQL LIKE Operator / String Matching	1	1527	Patients With a Condi
SQL Length Function (LENGTH/CHAR_LENGTH)	1	1683	Invalid Tweets (SQL)
SQL Logical Indexing (`AND` condition)	1	1757	Recyclable and Low F
SQL String Functions (CONCAT, LEFT) running directly on the database engine	1	2504	Concatenate the Nam
SQL UNION ALL (Unpivoting)	1	1795	Rearrange Products T
Semaphores	1	1117	Building H2O
Serialized Memory Wait (The closure for the addition waits twice)	1	2723	Add Two Promises
Set Size vs. Half-Length Comparison	1	575	Distribute Candies
Shunting Yard Algorithm / Two Stacks (Operands & Operators)	1	3749	Evaluate Valid Expres
Simple Iteration (Small Constraints)	1	3162	Find the Number of G
Simple Linear Scan	1	2942	Find Words Containin
Simple WHERE clause with OR	1	595	Big Countries (SQL)
Single Pass	1	2210	Count Hills and Valley
Single Pass (In-Place)	1	83	Remove Duplicates fr
Single Pass (Shared Edge) Count land cells x 4. Subtract 2 for every shared edge (horizontal or vertical neighbor)	1	463	Island Perimeter
Single Pass (Space Detection)	1	2490	Circular Sentence
Single Pass / Linear Indexing	1	1572	Matrix Diagonal Sum
Single Pass / Offset Comparison	1	1566	Detect Pattern of Len
Single Pass / State Tracking (Current Streak)	1	2414	Length of the Longes
Single Pass Bound Check. O(N)	1	3452	Sum of Good Number
Single Pass Comparison (Pythagorean Theorem)	1	3000	Maximum Area of Lor
Single Pass Contraction Check	1	1752	Check if Array Is Sort
Single Pass Count Iterate through characters. Count a segment whenever you see a non-space character that follows a space (or is at index 0)	1	434	Number of Segments
Single Pass First-Occurrence Map	1	2399	Check Distances Betw
Single Pass Linear Scan	1	1629	Slowest Key
Single Pass Max Tracking	1	747	Largest Number At Le
Single Pass Minimum Tracking	1	1779	Find Nearest Point Th
Single Pass State Toggle	1	2315	Count Asterisks
Single Pass State Tracking	1	551	Student Attendance R
Single Pass Streaming Maximum (King of the Hill)	1	1725	Number Of Rectangle
Single Pass with Lookahead (In-Place)	1	2816	Double a Number Rep
Single Pass with Running Minimum (Prefix Min)	1	2016	Maximum Difference
Single Pass with Three-Node Window	1	2058	Find the Minimum an
Single Pointer Iteration (count spaces and substring once)	1	1816	Truncate Sentence
Single-Pass Array Mapping	1	3173	Bitwise OR of Adjacer
Single-Pass Constant Space Scan	1	3019	Number of Changing

Single-Pass Early Exit	1	2057	Smallest Index With
Single-Pass Filtering	1	3065	Minimum Operations
Single-Pass Index Mapping	1	1929	Concatenation of Arra
Single-Pass Index Traversal (Step K)	1	3271	Hash Divided String
Single-Pass Level Accumulation	1	364	Nested List Weight Su
Single-Pass Max & Second Max Tracking	1	3833	Count Dominant Indi
Single-Pass Parity Check	1	3151	Special Array I
Single-Pass Pointer Distance	1	1437	Check If All 1's Are a
Single-Pass Pointer Scan	1	1678	Goal Parser Interpret
Single-Pass Range Intersection	1	1450	Number of Students
Single-Pass Reverse Scan (Right-to-Left Pointer)	1	1309	Decrypt String from
Single-Pass Scalar Tracking	1	1796	Second Largest Digit
Single-Pivot Circularity Check	1	2855	Minimum Right Shifts
Single-pass: Find the first and last occurrence of different words. The max distance is either LastIndex - FirstIndex or LastIndex - SecondFirstIndex	1	3696	Maximum Distance B
Sliding Boundary Pointers	1	2232	Minimize Result by A
SlowithFast Pointers	1	234	Palindrome Linked Lis
Square Root Decomposition (Block separation)	1	1714	Sum Of Special Even
Square Root Divisor Hunt	1	1390	Four Divisors
Square-Free Part Grouping	1	2862	Maximum Element-S
Standard Library formatting (bin/format)	1	3280	Convert Date to Bina
Standardization via Pre-filtering	1	831	Masking Personal Inf
Start at Square Root Start W at lfloor√{N}rfloor and decrement until N % W == 0. This ensures L and W are as close as possible	1	492	Construct the Rectan
State Array Snapshots (Drawing the entire array state for every microscopic time increment)	1	2751	Robot Collisions
State Tracking (Parity / Flip Count)	1	3141	Maximum Hamming
String Comparison (In-place `startsWith`)	1	2185	Counting Words With
String Comparison / Line Overlap	1	2446	Determine if Two Eve
String Doubling Trick Concatenate S+S. Remove the first and last characters. If S is still a substring of the result, it has a repeated pattern	1	459	Repeated Substring
String Functions	1	1543	Fix Product Name For
String Mutation Buffer (Visualizing repeated string recreations for each swap in languages with immutable strings)	1	2785	Sort Vowels in a Strin
String Predicate Filtering	1	3150	Invalid Tweets II (SQ
String Slicing / Cyclic Shift	1	3210	Find the Encrypted S
String Split	1	1859	Sorting the Sentence
Substring Slicing	1	2678	Number of Senior Cit
Suffix Array Traversal (Right-to-Left Scan)	1	3818	Minimum Prefix Rem
Suffix Sum of Shifts	1	848	Shifting Letters
Symmetric Construction (Pairing)	1	1304	Find N Unique Intege
Symmetry Grouping (2x2 Blocks	1	3240	Minimum Number of
Synchronization (Mutex/Locking)	1	1279	Traffic Light Controlle
Synchronization (Semaphores / Mutex	1	1195	Fizz Buzz Multithread
Targeted Filter	1	2230	The Users That Are E
Temporal State Snapshots (Visualizing the position of every robot at every tick of time)	1	2731	Movement of Robots
Temporary Object Accumulation (Creating massive amounts of intermediate merged objects)	1	2722	Join Two Arrays by ID
Temporary Property Allocation (Mutation of the target object)	1	2754	Bind Function to Con
The Circular Link (Ring)	1	61	Rotate List
The Reversal Algorithm	1	189	Rotate Array
Three-Phase Linear Scan	1	57	Insert Interval
Tokenization	1	1455	Check If a Word Occu
Top-3 Length Tracking per Character	1	2982	Find Longest Special
Top-3 Variables Keep `first`, `second`, `third` initialized to -Infinity. Update them sequentially as you iterate O(N)	1	414	Third Maximum Num
Trust Score Array (In-Degree vs Out-Degree)	1	997	Find the Town Judge
Two Heaps (Priority Queues) / TreeSet	1	1606	Find Servers That Ha
Two Passes (Prefix and Suffix) using Max-Heaps of size K	1	2519	Count the Number of
Two Queues (Enter and Exit) with State Tracking	1	2534	Time Taken to Cross
Two Queues (R_index, D_index)	1	649	Dota2 Senate
Two Stacks (Amortized O(1) - "Lazy Transfer")	1	232	Implement Queue us
Two Stacks (Predecessor & Successor)	1	272	Closest BST Value II
Two Stacks / Gap Buffer	1	2296	Design a Text Editor
Two Stacks Push all nodes of both lists onto two stacks. Pop from both to add from right-to-left	1	445	Add Two Numbers II
Two-Pass / Prefix Tracking	1	1422	Maximum Score Afte
Two-Pass Inclusion/Exclusion (At-Most K)	1	795	Number of Subarrays
Two-Pass Linear Traversal (Head-to-Tail)	1	3294	Convert Doubly Linke

Two-Pass Min-Sweep (Left-to-Right	1	821	Shortest Distance to
Two-Pass Monotonic Logic (Min/Max Tracking)	1	581	Shortest Unsorted Co
Two-Way Summation (Circular Array)	1	1184	Distance Between Bu
Two-way expansion from 'start' index	1	1848	Minimum Distance to
UNION of two filtered sets	1	1789	Primary Department
Update-History Log (Inverse Search)	1	1476	Subrectangle Queries
Vectorized Assignment: df['bonus'] = df['salary'] * 2	1	2881	Create a New Column
Visited Set / In-Place Marking	1	565	Array Nesting
WHERE clause filtering directly in SQL engine	1	1821	Find Customers With
Win-Condition Invariance Check	1	794	Valid Tic-Tac-Toe Stat
Wormhole Teleportation (O(N) Pre-matching)	1	1190	Reverse Substrings B
Z-Algorithm	1	2223	Sum of Scores of Bui
Z-Algorithm (Prefix & Suffix Matching)	1	3303	Find the Occurrence
Zeller's Congruence / Sum of Days	1	1185	Day of the Week
df.drop_duplicates(subset=['id'])	1	2882	Drop Duplicate Rows
df.dropna(subset=['name'])	1	2883	Drop Missing Data
df.head(3) (Slicing)	1	2879	Display the First Thre
df.melt(id_vars=['product'], ...)	1	2890	Reshape Data: Melt
df.pivot(index='month', columns='city', values='temperature')	1	2889	Reshape Data: Pivot
df.rename(columns={...})	1	2885	Rename Columns
df.shape Attribute Access	1	2878	Get the Size of a Data
df['grade'].astype(int)	1	2886	Change Data Type
df['quantity'].fillna(0)	1	2887	Fill Missing Data
pd.concat([df1, df2])	1	2888	Reshape Data: Conca
setTimeout	1	2821	Delay the Resolution
valueOf() and toString() Overriding	1	2695	Array Wrapper (JS)